Assessment 4 Final Project Report

Budgestic – AI-Powered Budgeting & Receipt Analysis App

Subject: IT3090
Assessment: Assessment 4 – Final Demonstration and Project Report
Project Title: Budgestic – AI-Powered Budgeting & Receipt Analysis App

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Student Information

Student ID	Student Name
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Project Supervisor/Advisor: Dr. Sherif Haggag

GitHub Repository Link

Additional resources such as the Figma application design link, README file, testing results, and documentation for users can be found at the following GitHub repository link:

GitHub Repository: https://github.com/Biraj1522/Budjestic.git

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1. Project Summary

Budgestic is a budgeting and receipt analysis mobile app that uses artificial intelligence. Budgestic allows users to analyze their personal finance by uploading or scanning their receipts, monitoring their incomes and expenditures, checking cash flow, and even analyzing spending habits using AI technology.

Objective of Project The primary objective of the project is to minimize the manual process of tracking expenditures and enable users to take better decisions regarding their finances. Most users find themselves losing their receipts, forgetting about their expenditures, and even making unnecessary expenditures without budgeting. The solution offered by Budgestic is through a user-friendly mobile application which tracks expenditures.

For Assessment 4, the project provides the entire design of the app, including the navigation process, design layout of the screens, functionality details, testing documentation, and user documentation. The design of the application is done using Figma, while future technical implementation will be done using Flutter, Tesseract OCR, SQLite, and lightweight AI/ML models.

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2. Project Objectives

The primary goals of the Budgestic project are as follows:

1. To develop an adaptive and intuitive budgeting app on mobile in Figma.
2. To enable uploading and scanning of receipts.
3. To extract details from receipts like item name, price, date, and total amount.
4. To track the income, expenses, and cash flow.
5. To offer artificial intelligence-based suggestions on expenditures.
6. To predict purchases made often.
7. Navigation flow of the mobile application.
8. Preparation of all required documents, including final report, testing document, user manual, and README file.

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3. Scope of the Project

3.1 In Scope

- Mobile application interface design.
- Interactive screen navigation.
- Receipt upload and scanning feature.
- OCR-based receipt data extraction concept.
- Expense tracking.
- Income and cash flow monitoring.
- Spending analysis.
- Basic financial reports and summary.
- Smart recommendation feature.
- Frequently purchased item prediction.
- User documentation and testing documentation.
- README and GitHub supporting files.

3.2 Out of Scope

- Bank API integration.
- Web-based version of the system.
- Advanced investment tools.
- Real-time financial institution syncing.
- Full commercial deployment.

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4. Target Users

Target users of Budgestic include:

- Students who require managing their day-to-day expenditure.
- Workers who wish to track their incomes and expenditures.
- Mobile phone users who are concerned about managing their personal finance.
- Mobile phone users who buy groceries regularly and wish to avoid unnecessary purchases.
- Mobile phone users who desire a budgeting application that is simple and can be used offline.

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5. Software Development Methodology

The development process was based on the agile model. This model was appropriate for this project since the project involved different phases such as planning, designing the application, testing, and improvement.

Sprint	Focus Area	Main Tasks
Sprint 1	User Interface Planning	Identify required screens and user flow
Sprint 2	App Design	Design welcome, login, dashboard, and navigation screens
Sprint 3	Receipt and Expense Screens	Design receipt upload and expense tracking screens
Sprint 4	Reports and Insights Screens	Design cash flow, reports, and spending insight screens
Sprint 5	Prediction and Notification Screens	Design prediction, reminder, and notification screens
Sprint 6	Testing and Documentation	Test app navigation and prepare final documents

Agile helped the team work in smaller sections, review progress regularly, and make improvements based on feedback.

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6. System Requirements

6.1 Functional Requirements

High Priority

- Users can view the app welcome screen.
- Users can navigate to login and create account screens.
- Users can access the dashboard screen.
- Users can access the receipt upload screen.
- Users can use the receipt scanning process.
- Users can access expense tracking screens.
- Users can view cash flow report screens.

Medium Priority

- The system categorises expenses.
- The system analyses spending habits.
- The system generates financial insights.
- The system displays reports and summaries.

Low Priority

- The system predicts frequently purchased items.
- The system sends notifications for products that may run out.
- The system provides smart shopping recommendations.

6.2 Non-Functional Requirements

Requirement	Description
Usability	The app should be simple and easy to navigate.
Performance	The app should load and move between screens smoothly.
Security	The system should protect user financial data.
Privacy	The system should store user data locally where possible.
Scalability	The design should support future feature expansion.
Accessibility	The interface should use readable text and clear navigation.
Reliability	The system should store expense data accurately.
Maintainability	The design and documentation should be clear for future development.

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7. System Design

7.1 Application Architecture

The Budgestic application operates based on the rules of operation of a mobile app. First, the user reaches the welcome page from which he/she either logs into the application or registers to go to the dashboard page. At this stage, the user has access to the functionality provided by the app, which includes uploading receipts, managing expenses, cash flow, analyzing expenditure, forecasting, getting notifications, and settings.

7.2 System Architecture

The whole system relies on the architecture of the mobile app. The user uploads his/her receipts through the mobile app and receives financial advice. The OCR module takes out the necessary data from the receipts. The collected information is then saved in the local SQLite database. After that, the AI/ML module does the analysis and makes recommendations.

7.3 Data Flow

- Step 1: The Budgestic application is launched on the user's mobile phone.
- Step 2: The user uploads or scans his/her receipt.
- Step 3: The OCR module reads the receipt.
- Step 4: Data cleaning and sorting.
- Step 5: Expenses' data are saved in the SQLite database.
- Step 6: Analysis of income and expenses.
- Step 7: Results presentation to the user.

7.4 Database Design

The system stores information such as:

- User details.
- Receipt records.
- Expense transactions.
- Income records.
- Categories.
- Spending insights.
- Prediction and notification data.

SQLite is used because it supports local storage, privacy, offline access, and lightweight mobile app development.

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8. UI/UX Design

The Budgestic interface was designed in Figma with a mobile-first approach. The app contains screens such as:

- Welcome screen.
- Login screen.
- Create account screen.
- Dashboard.
- Upload receipt screen.
- Expense tracking screen.
- Spending insights screen.
- Prediction screen.
- Cash flow report screen.
- Budget settings screen.
- Notifications screen.
- Profile and settings screen.

Design Principles

- **Simplicity:** Simple and clear design.
- **Consistency:** Consistent buttons, colors, icons, and spacing.
- **Accessibility:** Accessible texts and labels.
- **Mobile-first design:** Mobile-first screens.
- **Easy navigation:** Easy navigation among sections.
- **Visual hierarchy:** Financial information is displayed using visual hierarchy.
- **Alerts/feedback:** Status alerts and feedback are available.

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9. Implementation Summary

The implementation of Assessment 4 consisted of the complete design of the mobile application using Figma. The mobile application will be the design, layout, navigation, and interface design of Budgestic.

Tool / Technology	Purpose
Figma	Mobile app design and UI/UX

Tool / Technology	Purpose
GitHub	Repository for documentation and supporting files
Flutter	Mobile application development
Dart	Programming language for Flutter
Tesseract OCR	Receipt text extraction
SQLite	Local database storage
Python	AI/ML model development
Android Studio / VS Code	Development and debugging tools

The app was designed to support budgeting, receipt tracking, spending analysis, and prediction features.

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10. Testing Summary

Tests were run to test the navigation, screen flow, layout, and usability of the app. Some of the tests run include:

- App navigation tests.
- Layout tests of the screen.
- Button and links tests.
- User flow tests.
- Testing the receipt upload screen.
- Testing the expense tracking screen.
- Testing the report and insight screen.
- Usability tests.
- Visual consistency tests.

A separate testing summary file is available and has been uploaded on GitHub.

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11. User Documentation Summary

A separate user documentation file has been created and uploaded on GitHub explaining the way users can access and navigate through the Budgestic application design. The user documentation file consists of:

- How to access the app design.
- Login and account creation guide.
- Receipt upload guide.
- Expense tracking guide.
- Viewing reports and insights.
- Using prediction and notification screens.
- Troubleshooting information.
- App limitations and future development notes.

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12. Challenges Faced

Challenge	Explanation	Solution
App design limitation	The app is designed in Figma and does not have full backend functions.	Clearly document the current design and future implementation direction.
OCR functionality	Receipt scanning requires OCR integration.	Use Tesseract OCR in the planned technical implementation.
AI prediction	Spending insights and predictions require user data and AI/ML integration.	Use lightweight AI/ML models in future development.
Time management	Multiple screens and documents had to be completed.	Use sprint planning and divide tasks.
Team coordination	Group members needed to contribute regularly.	Use shared tools and GitHub for supporting files.

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13. Project Outcomes

Budgestic project outcomes are:

- Formation of a clearly defined idea of the mobile app.
- Creation of an entire Figma design of the mobile application.
- Determination of the main needs of the users.
- Recording of the functional and non-functional requirements.
- Creating of UI/UX screens and navigation.
- Recording of the system architecture and database design.
- Creation of testing documentation.
- Creation of user documentation.
- Creation of README for GitHub.
- Creation of the final report that covers all aspects of the project development process.

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14. Future Improvements

Potential future developments for Budgestic may include:

- Design and develop the application according to Figma design.
- Add an OCR receipt scanner.
- Add an SQLite database.
- Add AI-based spending analysis.
- Predict frequently purchased items.
- Improve user interaction and animations.
- Add bank API integration.
- Add a cloud backup feature.
- Add report generation in PDF/excel.
- Add budget goals.
- Add spending limit alerts.

- Release the app on Google Play/App store.

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15. Conclusion

Budgestic is a mobile application which assists its users in budget management and receipt analysis. This application proves the capability of its users to scan receipts, manage their expenses, cash flow, and receive financial counseling through the medium of a mobile application.

The intended features, user interface, navigation, and technical roadmap of the mobile application are provided here. It would be of immense benefit to students, working individuals, and other general users who want to utilize an easily accessible and privacy-focused budgeting app.

Here are the final report, testing report, user documentation, README file, and project files.

Appendix A: Table of Contributions

Student Name	ID Number	Contribution Percentage
Rajat Adhikari	H20937	33.33%
Biraj Thapa	ULG20814	33.33%
Bishwash Thapa	ULG20520	33.34%
Total		**100%**

Appendix B: GitHub Contents

The GitHub repository should include:

- Final project report.
- Figma app design link.
- README file.
- Testing summary.
- User documentation.
- Screenshots or UI designs.
- Any supporting diagrams.
- Project files and setup instructions.